Tayyba Abid

AI & Web Developer

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LinkedIn
Portfolio

PROFILE

AI & Web Developer skilled in building intelligent applications using Generative AI, RAG-based systems, and modern web frameworks. Experienced in developing full-stack solutions with Django, FastAPI, and Streamlit, and creating AI powered tools including coding assistants, quiz generators, and document-based chatbots. Proficient in Prompt Engineering, LangChain, and deploying production-ready systems. Passionate about leveraging AI to solve real-world problems with clean, scalable architectures.

EDUCATION

BS in Software Engineering — 2022 – 2026

COMSATS University Islamabad
CGPA: 3.53

PROJECTS

AI Learning Suite | FYP | 2024 – Present

FastAPI, Streamlit, LangChain, Prompt Engineering, Retrieval-Augmented Generation (RAG), python-docx, PyMuPDF, python-pptx, Vector Databases.

Two Streamlit applications consuming FastAPI endpoints:

Code Mentor Ai:

Adaptive AI tutor that detects user proficiency (beginner / intermediate / advanced), answers programming queries only, supports file uploads for context-aware Q&A, and auto-generates a Word quiz + answer key after every 6 queries.

Quiz Generator:

- Accepts Docs, PPT, and PDF uploads, chunks the text, and generates quizzes by difficulty (Easy / Medium / Hard) and format (MCQs / Short / Long). Includes an AI refinement prompt box and exports the final quiz as PDF, Docs, or PPT.

Multi-Bot Chat Base System

FastAPI, LangChain, Vector Databases, RAG, Python

- Built a SaaS-style chatbot platform similar to Chatbase.co where each user can upload documents and generate multiple isolated, document-aware chatbots each bot scoped strictly to its own data.
- Implemented per-user, per-bot vector stores ensuring complete data isolation across sessions.

Smart PDF Chat (RAG)

LangChain, RAG, Streamlit, Vector Databases, Python

- RAG-based PDF chat application where users upload documents and receive AI answers grounded strictly in the document content no hallucination beyond the provided source.
- Deployed with Streamlit for an intuitive, interactive user experience.

WordPress Web Development

WordPress, Bootstrap, Tailwind CSS, PHP

- POLOPOINT custom WordPress business site with responsive design.
- Al-Rafiq Mart e-commerce store with product management.
- Connect Intermodal corporate logistics website.

SKILLS & TECHNOLOGIES

Generative AI & LLMs: RAG-based Systems, LangChain, Prompt Engineering, Fine-Tuning, Hugging Face Transformers, Vector Databases, Chatbot Development

Natural Language Processing (NLP): Text Classification, Sentiment Analysis

Web Development: Django, FastAPI, Bootstrap, Tailwind CSS, WordPress, REST APIs, Streamlit

MLOps & Backend: Docker, FastAPI, REST APIs, Microservices Architecture, Production Pipelines

Programming Languages: Python, R, SQL, JavaScript, C++, Html.

Tools & Platforms: Git & GitHub, VS Code, Jupyter Notebook, Hugging Face, google colab.